Types Of Software Engineers

A Quick Handbook

To to get into any professions

By SHAILESH SHAKYA @BEGINNERSBLOG.ORG

1.Front-End Engineers

Role:

- Create the look and feel of websites, ensuring they are visually appealing and easy to navigate.
- Make sure interactive elements function as intended.

Tools:

Core Technologies:

- HTML (Hyper Text Markup Language)
- CSS (Cascading Style Sheets)
- JavaScript

Build Tools & Package Managers:

- Webpack
- Parcel
- Vite (fast development server)
- npm
- yarn

Frameworks & Libraries:

- React
- Angular
- Vue.js
- Svelte (gaining popularity for its performance benefits)

Styling & Design:

- Bootstrap
- Tailwind CSS
- Material UI
- Chakra UI

Continue...

Learning Resources:

Free:

- The Odin Project's foundations course
- FreeCodeCamp
- Scrimba (interactive learning platform)

Paid:

- Coursera: IBM Front-End Developer
- Coursera: Introduction to Web Development with HTML, CSS, JavaScript
- Udemy: "The Complete 2024 Web Development Bootcamp"
- Udacity: "Front End Web Developer Nanodegree"
- edX: Professional Certificate in Front-End Web Developer.
- edX: IBM: Developing Front End Apps with React

2. Back-End Engineers

Role:

- Handle the server-side logic, databases, and APIs that power websites and applications
- Ensure data is stored securely, retrieved efficiently, and processed seamlessly

Tools:

Server-Side Languages:

- Python (with Django or Flask)
- Java (with Spring Boot)
- Node.js (with Express.js)
- Ruby (with Ruby on Rails)
- PHP (with Laravel or Symfony)
- Go (for high-performance systems)

Databases:

- Relational Databases (SQL):
  - MySQL
  - PostgreSQL
- NoSQL Databases:
  - MongoDB
  - Cassandra
  - Firebase (Realtime Database)

Cloud Platforms:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

APIs & Web Services:

- REST (Representational State Transfer)
- GraphQL

Other Tools

- Git (version control)
- Docker (containerization)

Continue...

Learning Resources:

Free:

- The Odin Project's Node.js course
- FreeCodeCamp's Back End Development and APIs Certification
- "CS50's Web Programming with Python and JavaScript" by Harvard University (available on edX)

Paid:

- Coursera: "IBM Back-End Development
- Udemy: "The Complete Node.js Developer Course (3rd Edition)" or "Python and Django Full Stack Web Developer Bootcamp"
- Udacity: "Cloud Developer Nanodegree"

3. Full-Stack Engineers

Role:

- Possess expertise in both front-end and back-end development
- Work across the entire technology stack, handling everything from user interfaces to server-side logic

Tools:

Front-End:

- HTML, CSS, JavaScript
- Frameworks: React, Angular, Vue.js, Svelte
- Styling & Design: Bootstrap, Tailwind CSS, Material UI, Chakra UI
- Build Tools: Webpack, Parcel, Vite

Back-End:

- Server-Side Languages: Python, Java, Node.js, Ruby, PHP, Go
- Databases: MySQL, PostgreSQL, MongoDB, Cassandra, Firebase
- Frameworks: Django, Spring Boot, Express.js, Ruby on Rails, Laravel
- Cloud Platforms: AWS, Azure, GCP

Other Tools:

- Git (version control)
- Docker (containerization)
- APIs & Web Services (REST, GraphQL)

Continue...

Learning Resources:

Free:

- The Odin Project's full-stack curriculum
- FreeCodeCamp's Full Stack Development Certification

Paid:

- Coursera: IBM Full-Stack Developer
- edX: "Full Stack Development (IBM)
- Udemy: "The Complete 2023 Web Development Bootcamp"
- Udacity: "Full Stack Web Developer Nanodegree"

4. Mobile Engineers

Role:

- Specialize in creating applications for smartphones and tablets (iOS and Android)
- Design user-friendly interfaces, optimize performance, and ensure compatibility across various mobile platforms

Tools:

Native Development:

- iOS:
  - Swift (primary language)
  - Xcode (IDE)
  - SwiftUI (declarative UI framework)
- Android:
  - Kotlin (primary language)
  - Java (still widely used)
  - Android Studio (IDE)
  - Jetpack Compose (declarative UI framework)

Cross-Platform Frameworks:

- React Native (uses JavaScript)
- Flutter (uses Dart)

Other Tools

- Git (version control)
- Firebase (backend services, analytics, etc.)
- Testing frameworks (Espresso for Android, XCTest for iOS)

Continue...

Learning Resources:

Free:

- Official Documentation: Android Developer website, Apple Developer website
- "CS50's Mobile App Development with React Native" by Harvard University (available on edX)

Paid:

- Coursera: "Build Your First Android App (Project-Centered Course)" by CentraleSupélec
- Udemy: "The Complete iOS & swift App Development Bootcamp" or "The Complete Android 14 & Kotlin Development Masterclass"
- Udacity: "Android Basics Nanodegree" or "iOS Developer Nanodegree"

5. DevOps Engineers

Role:

- Bridge the gap between development, operations, and IT teams
- Automate processes, build CI/CD pipelines, and ensure seamless software delivery and deployment

Tools:

Containerization and Orchestration
- Docker
- Kubernetes
- Container registries (Docker Hub, Amazon ECR, Google Container Registry)

Infrastructure as Code (IaC):
- Terraform
- Ansible
- CloudFormation (AWS)
- Azure Resource Manager (ARM) templates

Cloud Platforms:
- AWS
- Azure
- GCP

Monitoring and Logging:
- Prometheus
- Grafana
- ELK Stack (Elasticsearch, Logstash, Kibana)
- Datadog
- New Relic

CI/CD Tools:
- Jenkins
- GitLab CI/CD
- CircleCI
- GitHub Actions
- Bamboo

Other Tools
- Git (version control)
- Scripting languages (Bash, Python)
- Configuration management tools (Chef, Puppet)

Continue...

Learning Resources:

Free:

- "Introduction to DevOps" on Coursera by IBM
- Official documentation and tutorials for Docker, Kubernetes, and Terraform
- DevOps blogs and communities (e.g., DevOps.com)

Paid:

- Coursera: "Architecting with Google Kubernetes Engine" by Google Cloud
- edX: "DevOps MicroMasters Program" by The Linux Foundation
- Udemy: "Docker and Kubernetes: The Complete Guide" or "Terraform - Complete Guide"
- Udacity: "Cloud DevOps Engineer Nanodegree"
- Certifications: AWS Certified DevOps Engineer - Professional

6. Data Engineers

Role:

- Design, build, and maintain the infrastructure for collecting, storing, and processing vast amounts of data.
- Ensure data is accessible and usable for analysis and decision-making.

Tools:

Data Pipelines & Orchestration:

- Apache Kafka (distributed streaming platform)
- Apache Airflow (workflow management)
- Dagster (data orchestration)

Data Warehousing and Lakes

- Snowflake (cloud-based data warehouse)
- Amazon Redshift
- Google BigQuery
- Apache Hive (data warehouse infrastructure on Hadoop)
- Delta Lake (open-source storage layer for reliability)

Cloud Data Platforms:

- AWS Glue (extract, transform, and load - ETL - service)
- Azure Data Factory
- Google Cloud Dataflow

Other Tools

- Python
- SQL

Databases

- Relational Databases (SQL) - PostgreSQL, MySQL
- NoSQL Databases - MongoDB, Cassandra

Continue...

Learning Resources:

Free:

- DataCamp's free introductory courses on data engineering and SQL
- Official documentation and tutorials for Apache Hadoop, Spark, Kafka, and Airflow
- Kaggle (data science and machine learning platform with datasets and competitions)

Paid:

- Coursera: "Google Cloud Big Data and Machine Learning Fundamentals" by Google Cloud
- Udemy: "The Ultimate Hands-On Hadoop - Tame your Big Data!" or "Apache Spark with Scala - Hands On with Big Data!"
- Udacity: "Data Engineer Nanodegree"
- Databricks Academy: Courses on Apache Spark and Delta Lake

7. Security Engineers

Role:

- Identify and mitigate vulnerabilities in systems and applications
- Design and implement security measures to protect against cyber threats and data breaches

Tools:

Vulnerability Assessment & Penetration Testing:

- Nessus, OpenVAS (vulnerability scanners)
- Metasploit, Burp Suite (penetration testing tools)
- OWASP ZAP (open-source web application security scanner)

Encryption & Cryptography:

- OpenSSL
- GPG

Other Tools

- Kali Linux (security-focused operating system)
- Security Orchestration, Automation, and Response (SOAR) platforms

Network Security:

- Firewalls (e.g., pfSense, Cisco ASA)
- Intrusion Detection/Prevention Systems (IDS/IPS) (e.g., Snort)
- Wireshark (network protocol analyzer)

Cloud Security:

- Cloud-specific security tools and services (e.g., AWS Security Hub, Azure Security Center)

Security Information and Event Management (SIEM):

- Splunk
- IBM QRadar
- Elastic SIEM

Continue...

Learning Resources:

Free:

- Open Web Application Security Project (OWASP) resources and guides
- "Introduction to Cyber Attacks" on Coursera by New York University
- Cybrary (online cybersecurity training platform)

Paid:

- Coursera: "Introduction to Cybersecurity" by University of Washington or "Security Engineering" by Arizona State University
- edX: "Cybersecurity Fundamentals" by RIT or "Introduction to Cloud Security" by IBM
- Udemy: "The Complete Cyber Security Course : Hackers Exposed!"
- SANS Institute: In-depth cybersecurity training and certifications
- Offensive Security: Penetration testing and ethical hacking courses (e.g., OSCP certification)

By SHAILESH SHAKYA @BEGINNERSBLOG.ORG

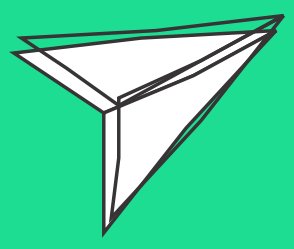
Created by Shailesh Shakya

@BEGINNERSBLOG.ORG

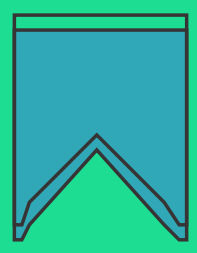
Did you find this post helpful?
Please...

LIKE

COMMENT



REPOST



SAVE